

Anomaly Detection based on Broad Learning System for Rolling Element Bearing Fault Diagnosis

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ABSTRACT

The latest proposed Broad Learning System (BLS) demonstrates an efficient and effective learning capability in many machine learning problems. In this paper, we develop a BLS based anomaly detection method (AD-BLS) for rolling element bearing fault diagnosis. Our method applies wavelet packet transformation (WPT) to extract the energy features, which will be further fed to the proposed BLS-based model to learn the intrinsic representations for the normal states of rolling bearings to conduct anomaly detection. Benefiting from the WPT pre-process, AD-BLS can be effectively trained with a small size of data. Moreover, due to the online updating mechanism of BLS, the designed AD-BLS can be easily updated with new extra data without re-design or retraining from scratch, which largely improves the flexibility of our method in real-world applications. Experimental results on the rolling element bearing fault diagnosis dataset further demonstrate the effectiveness of our method.

CCS CONCEPTS

• Computing methodologies → Machine learning; • Applied computing → Physical sciences and engineering.

KEYWORDS

Anomaly detection, broad learning system, fault diagnosis, single-layer feedforward neural network (SLFN)

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1 INTRODUCTION

Anomaly detection (AD) [4, 24, 28] is one of the most important applications among many smart city applications [6–8, 35], and have been widely applied in a variety of domains, including cybersecurity [2, 32], medical diagnosis [9, 14, 30], fault and damage detection [1, 20, 34]. Technically, AD methods are designed to model the patterns in normal data to recognize outliers, or in other words, atypical data. Such an imbalance data scenario naturally meets the fault diagnosis problem in the real-world industrial fields, where the fault samples are usually rare or even absent since the industrial systems are required to be as reliable as possible. Therefore, AD has shown its great practical significance in the health monitoring of unmanned aerial vehicles (UAV) [11], rotating machinery [16], etc.

The AD problem has been studied for a long time [4], while recent machine learning technologies, especially deep learning methods [24, 28] have started to play a more and more important role in this area [1, 16, 36, 39]. Although these deep learning based AD methods can usually achieve promising performance in many fault detection cases [5], the training procedure of the corresponding neural networks is conducted by the cumbersome backpropagation algorithm, which is time-consuming and easy to get stuck in local minima. However, single-layer feedforward neural networks (SLFN) based on random vector functional-link network (RVFLNN) [25] provide an efficient way for network training: using the Moore-Penrose generalized inverse to calculate the tunable network parameters. With sophisticated designs, these networks can also have promising generalization ability even with only single-layer architectures, which can be faster than deep networks. Therefore, due to the fast training and inference speed, recent studies also show interest in building AD methods based on different SLFNs [10, 22, 26].

The latest proposed SLFN, Broad Learning System (BLS) [5, 12], shows an efficient and effective learning capability, where the original inputs are transferred to feature nodes and enhancement nodes by using randomly generated first-layer weights. Then the tunable output weights can be calculated easily by the Moore-Penrose inverse efficiently in a close form. Compared with other SLFN (such as extreme learning machine [13]), BLS has two advantages when applied in fault diagnosis scenarios. First, BLS has a stronger representational capability since hidden features are enhanced to learn the intrinsic features of the anomaly patterns from the original

inputs. Moreover, due to the incremental model update mechanism of BLS, the BLS based AD methods can be flexibly updated by the new data in the extra streaming training data without re-design or retraining from scratch, which consistently improves the AD performance of our method in real-world applications.

Therefore, in this paper, we propose a novel BLS based anomaly detection method (AD-BLS) for rolling element bearing fault diagnosis. Our method pre-processes the measured vibration data using discrete wavelet packet transform (WPT) to reduce the redundancy and extract the intrinsic features in the original input, which ensures that the proposed AD-BLS can be trained effectively even with limited training samples. Our contribution can be summarized as follow:

- A novel anomaly detection method based on BLS (AD-BLS) together with a WPT-based input pre-processing algorithm is developed in this paper. Moreover, we found that the proposed WPT pre-processing method can also improve the performance of other SLFN based AD methods.
- The AD-BLS can be easily updated by the new streaming data from the test sets without retraining from scratch, which means that the performance of our AD-BLS will be improved consistently during fault detection scenarios.
- Experimental and simulation results show that the proposed method can achieve promising performance in rolling element bearing fault diagnosis tasks compared to other neural network based or SLFN based AD methods.

2 RELATED WORK

2.1 Anomaly Detection (AD)

Recent studies have shown that the data-driven based AD methods can achieve promising results in many industrial applications. Due to the breakthroughs of deep learning methods in recent years, researchers have designed many deep neural network based AD methods [21, 29] and applied them in various engineering applications [1, 16, 33, 36, 39]. However, these deep learning based AD methods need a large number of data for network training, which is impractical for many fault detection applications. Moreover, the training and inference process of deep learning based AD can be time-consuming during applications. While the SLFN based AD methods have shown their strong ability in many AD tasks. Lv et. al developed an AD method for network intrusion based on an optimal hybrid kernel ELM [22]. An extreme learning machine based on mutual information (ELM-MI) is further proposed in [26] for change point detection. The SLFN for predictive maintenance in [3] can achieve similar performance compared to other neural network based methods with significantly fewer training samples.

2.2 Broad Learning System (BLS)

BLS is first proposed in [5] and has already been improved to address many different problems [12], such as semi-supervised learning[40], transfer learning[38], and uncertainty data modeling [15], etc. Moreover, BLS has also been exploited in various real-world applications, like network intrusions detection [18, 19, 37] and fault diagnosis [27]. Although the works in [27, 37] also use BLS to conduct anomaly detection, our method differs from them in two aspects:

- 1) AD-BLS is developed to conduct anomaly detection using time-series data for fault diagnosis.
- 2) The designed WPT based data pre-processing method can largely reduce the dimension of the inputs and can improve the performance of the proposed AD-BLS.

2.3 Preliminaries

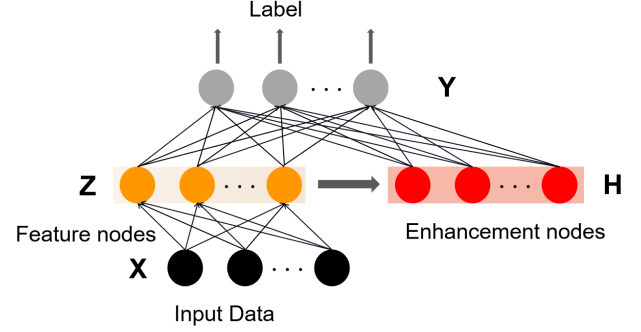


Figure 1: Illustration of Broad Learning System.

Let $\{X, Y\} = \{x_i, y_i\}_{i=1}^N$ be a dataset with N samples, where $x_i \in \mathbb{R}^{1 \times D}$ and $y_i \in \mathbb{R}^{1 \times K}$ are the inputs and labels (one-hot), respectively, where K is the number of classes. BLS firstly maps the inputs X to become the i -th mapped features Z_i by

$$Z_i = \phi_i(XW_{e_i} + b_{e_i}), \quad (1)$$

where W_{e_i} and b_{e_i} are the random weights with the proper dimensions and $\phi(\cdot)$ is the activation function. Then $Z^i \equiv [Z_1, \dots, Z_i]$ can be obtained by concatenating all the first i groups of mapping features. Similarly, the j -th group of enhancement nodes H_j , can be calculated using

$$H_j = \xi_j(Z^i W_{h_j} + b_{h_j}), \quad (2)$$

where W_{h_i} and b_{h_i} are the random generated and $\xi(\cdot)$ is the activation function. Then H_j can be denoted as $H^j \equiv [H_1, \dots, H_j]$.

In practice, the number of i and j can be selected depending upon the complexity of the modeling tasks. Assume we have n feature mappings and m groups of enhancement features with each feature mapping and enhancement features generating p nodes and q nodes, respectively. Then the obtained BLS features can be represented as

$$A_m^n = [Z^n \mid H^m] \in \mathbb{R}^{N \times (np+mq)}. \quad (3)$$

BLS aims to learn the best β which minimizes the sum of the squared losses of the prediction errors, which leads to the following problem

$$\min_{\beta} J_{\text{BLS}} = \frac{C}{2} \|\beta\|^2 + \frac{1}{2} \|Y - A\beta\|^2, \quad (4)$$

where the first term in (4) is the regularization term to prevent the overfitting and $\beta \in \mathbb{R}^{(np+mq) \times K}$. The optimal solution can be calculated efficiently by

$$\beta_{\text{BLS}}^* = (CI + A^T A)^{-1} A^T Y \quad (5)$$

where, I is the identical matrix.

The weights of BLS can be further updated easily without an entire training cycle when N_a new training samples $\mathbf{X}_a \in \mathbb{R}^{N_a \times D}$ are added. We can calculate the new BLS features $\mathbf{A}_x$ by Eq.(1) and Eq.(2) with $\mathbf{X}_a$ as inputs. Then the new BLS feature can be obtained by

$${}^x\mathbf{A}_m^n = \begin{bmatrix} \mathbf{A}_m^n \\ \mathbf{A}_x \end{bmatrix} \in \mathbb{R}^{(N+N_a) \times (np+mq)} \quad (6)$$

And the new optimal ${}^x\boldsymbol{\beta}^*$ can be calculated as

$${}^x\boldsymbol{\beta}^* = \boldsymbol{\beta}^* + (\mathbf{Y}_a^T - \mathbf{A}_x^T \boldsymbol{\beta}^*) \mathbf{B} \quad (7)$$

where

$$\mathbf{B}^T = \begin{cases} (\mathbf{C})^+, & \text{if } \mathbf{C} \neq 0 \\ (\mathbf{I} + \mathbf{D}^T \mathbf{D})^{-1} (\mathbf{A}_m^n)^+ \mathbf{D}, & \text{if } \mathbf{C} = 0 \end{cases}$$

and $\mathbf{D} = \mathbf{A}_x^T (\mathbf{A}_m^n)^+$, $\mathbf{C} = \mathbf{A}_x^T - \mathbf{D}^T \mathbf{A}_m^n$, $+$ means pseudo-inverse.

3 METHOD

In this section, we propose our AD method based on BLS for rolling bearing fault diagnosis. We first propose the WPT-based feature extraction method to generate wavelet packet energy features. Then the design of AD-BLS and how to incrementally update the existed AD-BLS with streaming training data are further introduced.

3.1 Pre-processing of the original data

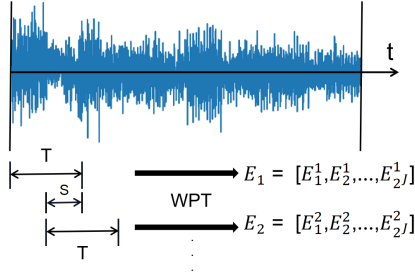


Figure 2: Extracting energy features by WPT from time series data. The time interval for calculating the energy features and the stride are denoted by T and S .

In industrial applications, the health condition of a rolling bearing is monitored by an accelerometer, which generates the time-series vibration data for bearing condition monitoring. While, the time time-series usually contains lots of redundant information, which can affect the AD performance of models. Therefore, we propose a data pre-processing method to extract the intrinsic features from the original time series samples by WPT and calculate the energy features at different sub-band frequencies as the input to conduct the subsequent anomaly detection. Such a pre-process procedure is illustrated in Fig. 2. Given a sub-band level J , T time points will be decomposed into 2^J sub-regions with the corresponding coefficients Ω_k^i . Then the wavelet packet energy of each sub-band frequency can be calculated as

$$e_i = \sum_{k=1}^K (\Omega_k^i)^2, \quad i = 1, \dots, 2^J, \quad K = \frac{T}{2^J}. \quad (8)$$

The energy at each frequency band is further normalized by $E_i = e_i / \sum_{j=1}^{2^J} e_j$. To this end, T time points are transformed into wavelet packet energy features, where the original feature dimension reduces from T to 2^J . Generally, $J = 4$ can work well in our experiments. Such a data pre-processing has two advantages: 1) The complexity of the AD-BLS will be lower due to the reduction of the input dimension (from T to 2^J , and $T > 2^J$). 2) Moreover, WPT ensures that the AD-BLS can be effectively trained with a limited number of training data.

3.2 AD-BLS

When the training data comes from only normal samples, we modify the BLS to learn patterns from the target samples (the normal states in anomaly detection problems) and reject all samples that are significantly different from the target class. Specifically, after we obtain the BLS features $\mathbf{A}$, the AD-BLS also aims to learn the best $\boldsymbol{\beta}$ which minimizes the following problem

$$\min_{\boldsymbol{\beta}} J_{\text{AD-BLS}} = \frac{C}{2} \|\boldsymbol{\beta}\|^2 + \frac{1}{2} \|\mathbf{T} - \mathbf{A}\boldsymbol{\beta}\|^2, \quad (9)$$

where the $\mathbf{T}$ is different from $\mathbf{Y}$ in Eq. (4). Compared to the classical one-class BLS [37], where the $\mathbf{T}$ is defined as $\mathbf{T} = [t, \dots, t]$, where $t \in \mathbb{R}$ and represents the target class, the proposed AD-BLS defined $\mathbf{T}$ as $\mathbf{T} = [t, \dots, t]$, where $t \in \mathbb{R}^{S \times 1}$. Actually, the label t of target class can be arbitrary except 0. Generally, S can be set as 1. After we obtain the optimal $\boldsymbol{\beta}^*$ using Eq. (5), the target distances of each training samples can be calculated by

$$\Phi(i) = \|\mathbf{A}(\mathbf{x}_i)\boldsymbol{\beta}^* - \mathbf{T}\|_2. \quad (10)$$

After we sort all $\Phi(1), \dots, \Phi(N)$, where $\Phi_1 \leq \Phi_2 \leq \dots \leq \Phi_N$, we can use a percentage $\eta \in [0, 1]$ to select a reject threshold: $\zeta = \Phi(\lfloor \eta N \rfloor)$ to conduct the anomaly detection. With a new sample $\hat{\mathbf{x}}$, we first calculate the corresponding target distance $\Phi(\hat{\mathbf{x}}) = \|\mathbf{A}(\hat{\mathbf{x}})\boldsymbol{\beta}^* - \mathbf{T}\|_2$, and identify $\hat{\mathbf{x}}$ as an outlier if $\Phi(\hat{\mathbf{x}}) > \zeta$.

3.3 Incremental learning with streaming data

After we obtain the initial AD-BLS, the model can be further updated incrementally by the new streaming data from the monitoring system using the incremental update method by Eq.(6) and Eq.(7). In real-world applications, we collect a small number of training samples under the health condition and train the AD-BLS with the aforementioned procedure. Next, when there are new training data, we can use them to conduct the online learning process for AD-BLS to incrementally update the model to include the new information. During the test, the trained model is fed by the observed sample to detect whether the bearing is in a healthy condition or not. We provide the complete procedure for AD-BLS based fault diagnosis scheme for rolling bearings in Fig. 3.

4 EXPERIMENTS

In this section, the proposed method and other related approaches are evaluated on the CWRU bearing dataset [31] from the Case Western Reserve University Bearing Data Center¹. A simulation study is also provided based on CWRU bearing dataset. Analytic experiments further explore more properties of our method. In all

¹<http://csegroups.case.edu/bearingdatacenter/home>

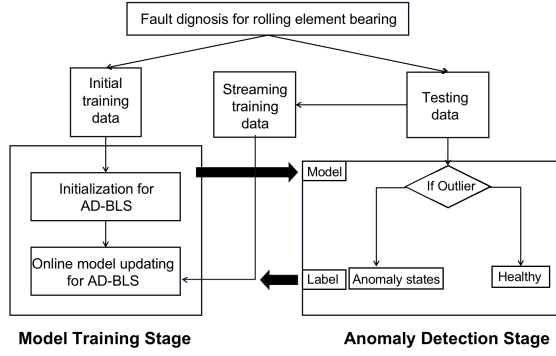


Figure 3: Flowchart of AD-BLS based anomaly detection for rolling element bearing.

the experiments below, the testing hardware and software conditions are as follows: Windows 10, Python 3.9, Intel-i5 2.4GHz CPU, 8G DDR3 RAM.

4.1 Data description

The CWRU dataset contains the vibration data collected by an accelerometer with a 12kHz sampling rate labeled with four health conditions of a running rolling bearing: (1) normal condition, (2) inner race fault, (3) outer race fault and (4) ball fault. By setting T and S , we can generate the samples of different states from the original time series data. For AD-BLS, we apply the WPT pre-processing method described in Sec.3.1 to generate the energy features. Empirically, we use $J = 4$ in our experiments.

4.2 AD performance on CWRU bearing dataset

4.2.1 Experimental setting. We first introduce the experimental settings for AD with CWRU dataset. We first randomly select 1000 samples from the normal condition to train the AD methods. Then we randomly select 200 samples from each state as the test set, where all three fault conditions are labeled as an anomaly state. T, S are set as 500, 250, respectively. Following previous researchers [26, 37], we consider four widely-used metrics for evaluation, including the accuracy (ACC), Precision (P), Recall (R), and F1-score (F1). All experiments are conducted 5 times and provided by the means and standard deviations of the results for each method. In our experiments, one-class ELM (OC-ELM) [17] and one-class BLS [37], one-class neural networks [23] are selected as baseline models. We further apply WPT to OC-ELM, which is denoted as OC-ELM* in our experiments.

In real-world applications, the data might not be measured under ideal conditions. To test the robustness of different methods, we further add Gaussian noise to the original time series data to simulate the measurement error during AD tasks. The noisy data is sampled from a Gaussian distribution, $N(0, \delta)$, where δ is the noise-level and means the variance.

4.2.2 Anomaly detection results. The results are shown in Table 1, where the highest values are marked by **bold**. From the results, we see that our method generally has better performance and lower

Table 1: Anomaly detection results with different methods (mean $\pm$ std) on CWRU bearing dataset. * means using the WPT features.

δ	method	ACC	Precision	Recall	F1-score
0	AD-BLS	99.9 $\pm$ 0.0	100.0 $\pm$ 0.0	99.8 $\pm$ 0.0	99.9 $\pm$ 0.0
	OC-BLS	91.1 $\pm$ 0.0	91.4 $\pm$ 0.0	97.3 $\pm$ 0.0	94.3 $\pm$ 0.0
	OC-ELM*	99.9 $\pm$ 0.0	100.0 $\pm$ 0.0	99.8 $\pm$ 0.0	99.9 $\pm$ 0.0
	OC-ELM	86.3 $\pm$ 0.2	90.3 $\pm$ 0.3	88.6 $\pm$ 0.1	90.6 $\pm$ 0.1
	OC-NN*	99.7 $\pm$ 0.2	99.7 $\pm$ 0.2	99.8 $\pm$ 0.2	99.8 $\pm$ 0.2
0.2	AD-BLS	92.9 $\pm$ 0.0	93.6 $\pm$ 0.0	97.2 $\pm$ 0.0	95.3 $\pm$ 0.0
	OC-BLS	75.6 $\pm$ 0.0	75.9 $\pm$ 0.0	99.0 $\pm$ 0.0	85.9 $\pm$ 0.0
	OC-ELM*	89.0 $\pm$ 0.1	96.1 $\pm$ 0.1	89.1 $\pm$ 0.3	92.4 $\pm$ 0.1
	OC-ELM	75.2 $\pm$ 0.2	75.3 $\pm$ 0.2	99.7 $\pm$ 0.2	85.8 $\pm$ 0.1
	OC-NN*	92.3 $\pm$ 0.5	94.2 $\pm$ 0.4	94.4 $\pm$ 0.4	94.3 $\pm$ 0.4
0.4	AD-BLS	77.9 $\pm$ 0.0	78.2 $\pm$ 0.0	97.8 $\pm$ 0.0	86.9 $\pm$ 0.0
	OC-BLS	75.1 $\pm$ 0.0	75.1 $\pm$ 0.0	100.0 $\pm$ 0.0	85.8 $\pm$ 0.0
	OC-ELM*	75.2 $\pm$ 0.1	75.2 $\pm$ 0.2	99.8 $\pm$ 0.2	85.8 $\pm$ 0.2
	OC-ELM	75.2 $\pm$ 0.1	75.2 $\pm$ 0.2	99.8 $\pm$ 0.2	85.8 $\pm$ 0.2
	OC-NN*	76.3 $\pm$ 0.5	78.1 $\pm$ 0.4	88.4 $\pm$ 0.4	83.25 $\pm$ 0.4

variance compared to other methods. Notably, AD-BLS can accurately identify all abnormal states under ideal measurement conditions ($\delta = 0$). With a higher noisy-level, the performance of all AD methods will decrease, which meets our intuition. We also observe that AD-BLS and OC-ELM* are superior to their counterparts without WPT pre-processing, which demonstrates the importance of the pre-processing based on WPT when dealing with the time series data. All these results show the effectiveness of the proposed method in the AD task for rolling element bearing monitoring.

4.2.3 Incremental learning for AD-BLS with streaming. We further test how the performance changes when we use extra training data to conduct the incremental learning for AD-BLS. The data with $\delta = 0.2$ is used in this experiments. The initial model is trained using 10 samples, then we gradually fuse the information from the new training data into the original AD-BLS model with four stages. From the results in Tab. 2, we see that updating with extra new training samples can consistently improve the performance of AD-BLS.

Table 2: Incremental learning with extra streaming data for AD-BLS. (+N) means that there are N more samples added compared to the initial stage. NL represents that there are noisy-labeled samples during incremental learning.

	10	1st(+40)	2nd(+490)	3rd(+990)	4th(+1990)
AD-BLS	93.0	94.4	94.6	95.7	96.4
AD-BLS (NL)	93.0	94.4	92.8	94.6	95.8

4.2.4 Robustness to label noise during incremental learning. Usually, the extra streaming data for updating the AD-BLS should be checked by experts and then applied for model training. However, what if we directly use the predicted results as the pseudo label to update the model? Here, we test such cases by introducing the noisy-label for some samples, which are actually anomaly states but wrongly classified by AD-BLS as health conditions. Five

noisy-labeled samples are mistakenly used at the second updating stage (+ 490 samples). The test results are shown in Table 2, from which we see that the noisy-labeled sample can actually affect the performance of the AD-BLS. However, the performance gradually further increases with more appropriate healthy condition samples introduced during the incremental learning.

4.3 Fault diagnosis simulation

This section provides a simulation study to evaluate the proposed method in real-world fault diagnosis application for rolling bearing. We first manually synthesize a time series data (Fig. 4) containing four parts of three states: 1) the normal conditions; 2) anomaly states caused by ball faults; 3) anomaly states caused by the poor connections of the sensor. For Condition-2, we concatenate the time data from ball faults to the normal time series data among the time dimension. The data of Condition-3 is drawn from the Gaussian distribution to represent that the accelerometer occurs error and does not work appropriately. The numbers of time points for the four intervals are 5000, 2500, 2000 and 3000 respectively.

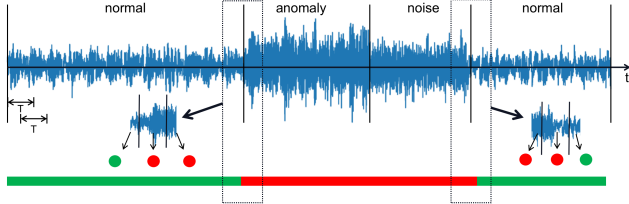


Figure 4: The fault diagnosis simulation. The simulated time series data is depicted by the blue wave. The AD-BLS conduct the AD task based on each sampled time series T and generated the results, which are denoted by green dots (healthy states) and red dots (anomaly states).

We use the synthesized time series to evaluate the proposed method, the detection results are shown at the bottom of Fig. 4. We see that our method will identify the abnormal states as long as the time interval T contains enough samples with anomaly states. In our simulation, 250 time points is enough for detecting the anomaly conditions. As the sampling frequency of the time series is 12kHz, and the inference latency of the proposed AD-BLS is around 3ms, the detection latency will be less than 23ms.

4.4 Analytic experiments

Regularization term (C) is selected from $\{10^{-11}, \dots, 10^{19}\}$. The Average Precision (AP, the area under the PR curve) on the test set using different C s are shown in Fig. 5 (a), from which we conclude that: The selection of C can affect the performance of the algorithm and therefore should be searched to guarantee the performance of AD-BLS. $C = 10^3$ usually shows the best performance.

Reject threshold ζ also affects the precision and recall of the final AD model. Here, with different ζ values, we obtain the PR-curve shown in Fig. 5 (b) of AD-BLS and OC-ELM*. By calculating the AP, we see that our method achieves better AP performance compared to ELM based AD method. Moreover, in real-world applications, we choose the ζ with the largest F1-score to conduct the AD tasks.

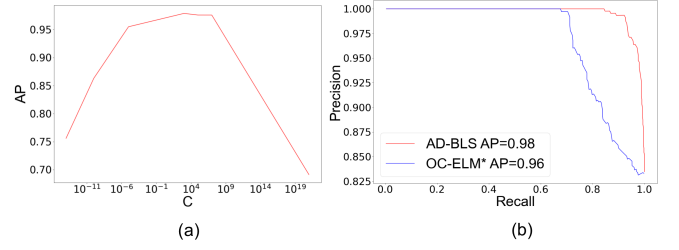


Figure 5: (a) Algorithm performance (average precision, AP) with the variation of different C . (b) The PR-curve for OC-ELM* and AD-BLS. The APs are also shown.

The time interval T and stride S are also important hyper-parameters. A small T with higher performance can ensure effective real-time AD during applications. However, the performance can be degenerated with too small T , since the information contained in each sample is not enough. We provide the AD results with different T ($S = T/2$) in Fig. 6, from which we can conclude that: 1) A small T can also achieve a promising performance with ideal measurement conditions ($\delta = 0$). 2) The performance of AD-BLS shows a significant improvement when we use a large T when $\delta = 0.2$. Therefore, to ensure the performance of our method we select $T = 500$ during applications.

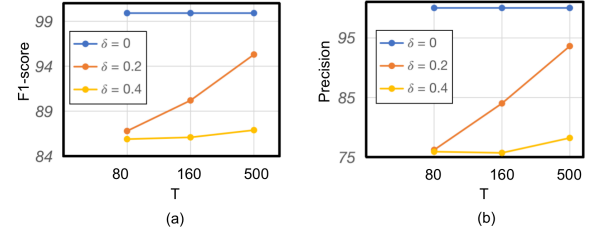


Figure 6: The performance of AD-BLS with different T and S under different noisy level. (a) F1 v.s. T . (b) Precision v.s. T .

5 CONCLUSION

This paper proposed a novel anomaly detection based on a broad learning system for rolling element Bearing Fault Diagnosis. Our method applies WPT to extract the energy features from the original time series data and uses BLS to conduct anomaly detection, which can accurately recognize the outlier running states for rolling element bearings. Moreover, the extra training samples can be easily applied with an incremental update method for AD-BLS. Experimental results demonstrate the effectiveness and efficiency of the proposed AD-BLS. Further works mainly include building sophisticated Bayesian learning-based methods, or minimax probability learning machines to further improve the performance of the BLS-based AD methods.

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